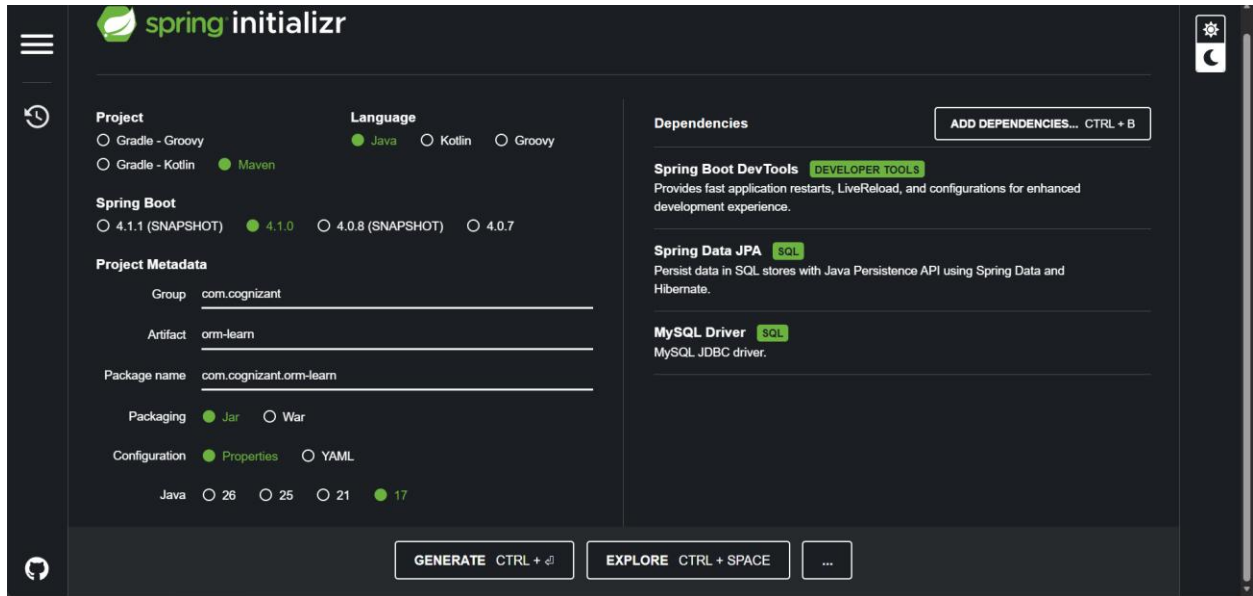


Creating the Eclipse Project using Spring Initializer:



The Spring Initializr web interface is shown in a dark theme. It features a sidebar on the left with navigation icons. The main content area is divided into three sections: Project, Language, and Dependencies. The Project section includes options for Project (Gradle - Groovy, Gradle - Kotlin, Maven), Spring Boot (4.1.1 (SNAPSHOT), 4.1.0, 4.0.8 (SNAPSHOT), 4.0.7), Project Metadata (Group, Artifact, Package name), Packaging (Jar, War), Configuration (Properties, YAML), and Java (Java, 26, 25, 21, 17). The Language section includes options for Language (Java, Kotlin, Groovy). The Dependencies section includes a list of dependencies (Spring Boot DevTools, Spring Data JPA, MySQL Driver) and an ADD DEPENDENCIES... button. At the bottom, there are buttons for GENERATE, EXPLORE, and a menu icon.

Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ Maven

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☒ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7

Project Metadata

Group

Artifact

Package name

Packaging ☒ Jar ☐ War

Configuration ☒ Properties ☐ YAML

Java ☐ Java ☐ 26 ☐ 25 ☐ 21 ☒ 17

Language

☒ Java ☐ Kotlin ☐ Groovy

Dependencies ADD DEPENDENCIES... CTRL + B

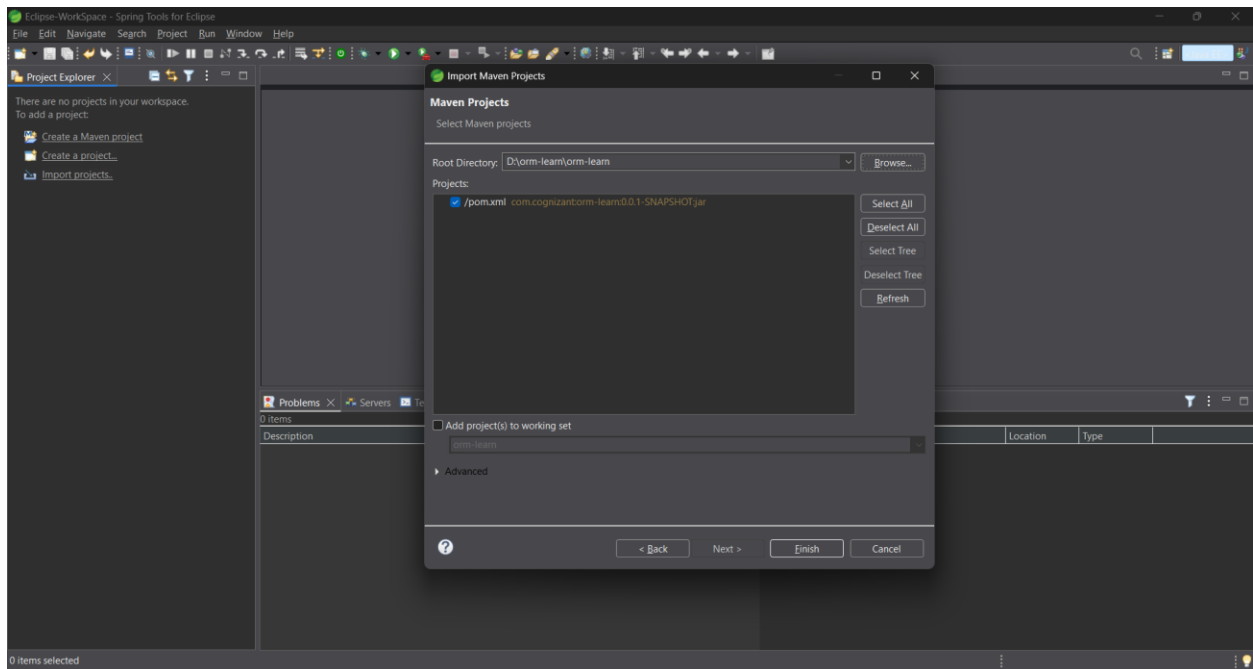
Spring Boot DevTools DEVELOPER TOOLS
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Data JPA SQL
Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.

MySQL Driver SQL
MySQL JDBC driver.

GENERATE CTRL + G EXPLORE CTRL + SPACE ...

Importing the Project using Spring Tool Suite:



Databases Creation:

```
MySQL 9.7 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.0 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

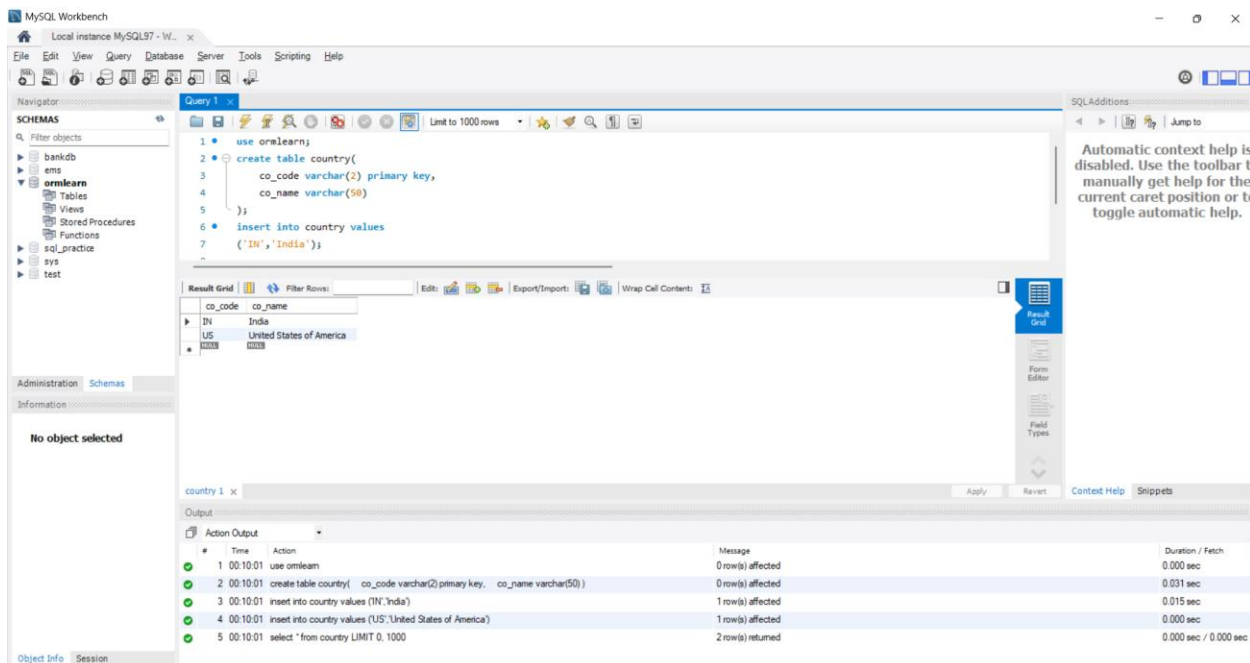
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database ormlearn;
Query OK, 1 row affected (0.406 sec)

mysql> |
```

Creation of Country table:



MySQL Workbench

Local instance MySQL57 - W...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

- Filter objects
- bankdb
- ems
- ormlearn
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sql_practice
- sys
- test

Query 1 x

```
1 use ormlearn;
2 create table country(
3     co_code varchar(2) primary key,
4     co_name varchar(50)
5 );
6 insert into country values
7 ('IN','India');
```

Result Grid

co_code	co_name
IN	India
US	United States of America

country 1 x

Output

#	Time	Action	Message	Duration / Fetch
1	00:10:01	use ormlearn	0 row(s) affected	0.000 sec
2	00:10:01	create table country(co_code varchar(2) primary key, co_name varchar(50))	0 row(s) affected	0.031 sec
3	00:10:01	insert into country values ('IN','India')	1 row(s) affected	0.015 sec
4	00:10:01	insert into country values ('US','United States of America')	1 row(s) affected	0.000 sec
5	00:10:01	select * from country LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Verification of Main Class:

Code:

```
package com.cognizant.orm_learn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);

    public static void main(String[] args) {

        SpringApplication.run(OrmLearnApplication.class, args);

        LOGGER.info("Inside main");

    }
}
```

Output:

[illegible]

```
21-07-26 00:19:02.790 restartedMain      INFO toryConfigurationDelegate
registerRepositoriesIn 213 Finished Spring Data repository scanning in 45 ms. Found
1 JPA repository interface.
21-07-26 00:19:03.079 restartedMain      INFO org.hibernate.orm.jpa
logPersistenceUnitInformation 102 HHH008540: Processing PersistenceUnitInfo [name:
default]
21-07-26 00:19:03.138 restartedMain      INFO org.hibernate.orm.core
logVersion 41 HHH000001: Hibernate ORM core version 7.4.1.Final
21-07-26 00:19:03.495 restartedMain      INFO SpringPersistenceUnitInfo
addTransformer 129 No LoadTimeWeaver setup: ignoring JPA class transformer
21-07-26 00:19:03.521 restartedMain      INFO c.z.h.HikariDataSource
getConnection 109 HikariPool-1 - Starting...
21-07-26 00:19:03.784 restartedMain      INFO c.z.h.pool.HikariPool
checkFailFast 580 HikariPool-1 - Added connection
com.mysql.cj.jdbc.ConnectionImpl@5cd97735
21-07-26 00:19:03.786 restartedMain      INFO c.z.h.HikariDataSource
getConnection 122 HikariPool-1 - Start completed.
21-07-26 00:19:03.870 restartedMain      INFO o.h.o.connections.pooling
logConnectionInfo 179 HHH10001005: Database info:
    Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
    Database driver: MySQL Connector/J
    Database dialect: MySQLDialect
    Database version: 9.7
    Default catalog/schema: ormlearn/undefined
    Autocommit mode: undefined/unknown
    Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
    JDBC fetch size: none
    Pool: DataSourceConnectionProvider
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown
21-07-26 00:19:04.553 restartedMain      in      INFO org.hibernate.orm.core
initiateService 53 HHH000489: No JTA platform available (set
'hibernate.transaction.jta.platform' to enable JTA platform integration)
21-07-26 00:19:04.598 restartedMain      INFO rEntityManagerFactoryBean
buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for
persistence unit 'default'
21-07-26 00:19:04.667 restartedMain      INFO .q.QueryEnhancerFactories
<clinit> 48 Hibernate is in classpath; If applicable, HQL parser will be used.
21-07-26 00:19:04.897 restartedMain      INFO c.c.o.OrmLearnApplication
logStarted 60 Started OrmLearnApplication in 2.956 seconds (process running for
3.527)
21-07-26 00:19:04.900 restartedMain      INFO c.c.o.OrmLearnApplication
main 20 Inside main
21-07-26 00:19:04.906 licationShutdownHook INFO rEntityManagerFactoryBean
destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
21-07-26 00:19:04.910 licationShutdownHook INFO c.z.h.HikariDataSource
close 349 HikariPool-1 - Shutdown initiated...
21-07-26 00:19:04.916 licationShutdownHook INFO c.z.h.HikariDataSource
close 351 HikariPool-1 - Shutdown completed.
```

Using other Files are Model, Repository and Service

Country.java

```
package com.cognizant.orm_learn.model;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name="country")
public class Country {

    @Id
    @Column(name="co_code")
    private String code;

    @Column(name="co_name")
    private String name;

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}
```

CountryRepository.java

```
package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.cognizant.orm_learn.model.Country;

@Repository
public interface CountryRepository
    extends JpaRepository<Country,String>{

}
```

CountryService.java

```
package com.cognizant.orm_learn.service;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.repository.CountryRepository;

import jakarta.transaction.Transactional;

@Service
public class CountryService {

    @Autowired
    private CountryRepository countryRepository;

    @Transactional
    public List<Country> getAllCountries(){

        return countryRepository.findAll();

    }

}
```

OrmLearnApplication.java

```
package com.cognizant.orm_learn;

import java.util.List;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.service.CountryService;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(
                OrmLearnApplication.class, args);

        countryService =
            context.getBean(CountryService.class);

        testGetAllCountries();

    }

    private static void testGetAllCountries(){

        LOGGER.info("Start");
```

$$\}$$

Output:

[illegible]


```
21-07-26 00:24:54.267 restartedMain      INFO SpringPersistenceUnitInfo
addTransformer 129 No LoadTimeWeaver setup: ignoring JPA class transformer
21-07-26 00:24:54.309 restartedMain      INFO c.z.h.HikariDataSource
getConnection 109 HikariPool-1 - Starting...
21-07-26 00:24:54.694 restartedMain      INFO c.z.h.pool.HikariPool
checkFailFast 580 HikariPool-1 - Added connection
com.mysql.cj.jdbc.ConnectionImpl@2f172a27
21-07-26 00:24:54.697 restartedMain      INFO c.z.h.HikariDataSource
getConnection 122 HikariPool-1 - Start completed.
21-07-26 00:24:54.807 restartedMain      INFO o.h.o.connections.pooling
logConnectionInfo 179 HHH10001005: Database info:
    Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
    Database driver: MySQL Connector/J
    Database dialect: MySQLDialect
    Database version: 9.7
    Default catalog/schema: ormlearn/undefined
    Autocommit mode: undefined/unknown
    Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
    JDBC fetch size: none
    Pool: DataSourceConnectionProvider
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown
21-07-26 00:24:55.655 restartedMain      INFO org.hibernate.orm.core
initiateService 53 HHH000489: No JTA platform available (set
'hibernate.transaction.jta.platform' to enable JTA platform integration)
21-07-26 00:24:55.709 restartedMain      INFO rEntityManagerFactoryBean
buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for
persistence unit 'default'
21-07-26 00:24:55.789 restartedMain      INFO .q.QueryEnhancerFactories
<clinit> 48 Hibernate is in classpath; If applicable, HQL parser will be used.
21-07-26 00:24:56.016 restartedMain      INFO c.c.o.OrmLearnApplication
logStarted 60 Started OrmLearnApplication in 3.311 seconds (process running for
3.895)
21-07-26 00:24:56.020 restartedMain      INFO c.c.o.OrmLearnApplication
testGetAllCountries 53 Start
21-07-26 00:24:56.199 restartedMain      DEBUG org.hibernate.SQL
logStatement 126 select c1_0.co_code,c1_0.co_name from country c1_0
21-07-26 00:24:56.235 restartedMain      DEBUG c.c.o.OrmLearnApplication
testGetAllCountries 61 countries=[Country [code=IN, name=India], Country [code=US,
name=United States of America]]
21-07-26 00:24:56.235 restartedMain      INFO c.c.o.OrmLearnApplication
testGetAllCountries 64 End
21-07-26 00:24:56.249 licationShutdownHook INFO rEntityManagerFactoryBean
destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
21-07-26 00:24:56.253 licationShutdownHook INFO c.z.h.HikariDataSource
close 349 HikariPool-1 - Shutdown initiated...
21-07-26 00:24:56.262 licationShutdownHook INFO c.z.h.HikariDataSource
close 351 HikariPool-1 - Shutdown completed.
```

